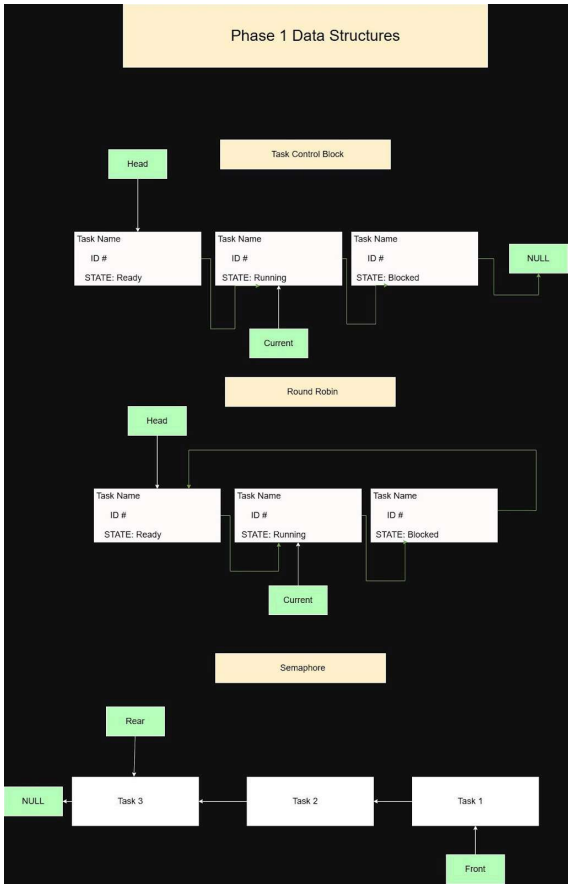
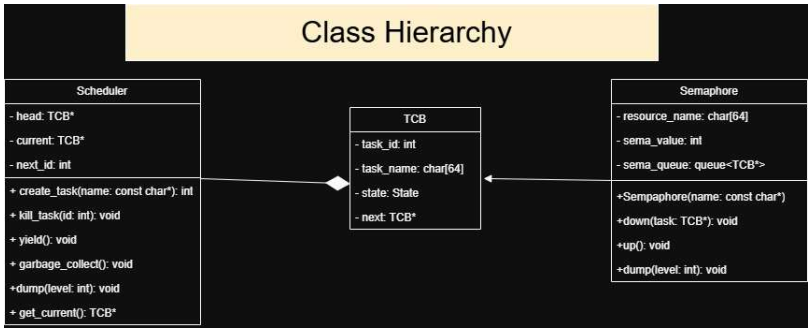
Design Diagrams:

A single linked list of TCBs is used to manage tasks within the scheduler. A pointer is in each TCB, pointing to the next task. This allows the scheduler to move, add, and delete tasks. There is a head pointer at the head of the list and a current pointer to the currently executing task, allowing round-robin scheduling. The semaphore uses a queue to block tasks when a shared resource is unavailable. Blocked tasks are stored in FIFO order. The scheduler and TCB has a composition relationship, as the TCB depends on the scheduler for creating and managing the tasks. The semaphore and TCB has an association relationship, as the semaphore references task, using the queue to control access.